Why mean-of-means fails

PARTITION A

[2, 4]

mean = 3

PARTITION B

[12]

mean = 12

(3 + 12) / 2 = 7.5 but (2 + 4 + 12) / 3 = 6

Original teaching example. No course materials used.

Keep a mergeable state

PARTITION A

[2, 4]

(sum, count) = (6, 2)

PARTITION B

[12]

(sum, count) = (12, 1)

Merge: (6, 2) + (12, 1) = (18, 3). Finalize: 18 / 3 = 6.

Original teaching example. No course materials used.